
Reading Positions Off a Genotype–Phenotype Model: A Negative Result on Attribution Reliability in a Multiplexed Splicing Assay

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Abstract

1 Reading the top attributed positions off a trained sequence–activity model is stan-
2 dard practice for guiding RNA therapeutic design [14], and that community has
3 argued such claims are validated anecdotally [22]. We test the practice on a real
4 multiplexed splicing assay and report that it is not reproducible. Models whose
5 held-out accuracy is statistically indistinguishable assign different top-3 positions
6 for 43% of held-out sequences at the best-fitting cell in our grid, rising to 94%
7 at the worst; no cell escapes it, so the failure is not confined to poorly-fitting
8 models. Tightening the accuracy match until 17 of every 18 surviving pairs are
9 discarded leaves that figure unchanged. Diagnosing why, we find three compound-
10 ing causes. The fitting objective does not identify the attribution, so the choice
11 among equally-accurate models is made by the initialization seed. Attribution
12 mass is concentrated — a median 98% of a sequence’s total in 3 of 9 positions at
13 the best cell — and we show in closed form that under such concentration a Spear-
14 man correlation of 0.71 is what *perfect* agreement about which positions matter
15 already produces, so the rank correlations normally used to check agreement can-
16 not distinguish success from failure. And the criterion the field selects on does not
17 constrain the function: tied models differ pointwise on the scale of the assay’s own
18 Poisson observation noise, so one or two held-out measurements separate them as
19 functions even though their aggregate accuracy is indistinguishable — the data
20 could choose, and the accuracy criterion does not. We also report two ways our
21 own validation nearly hid the result: a control that read as passing because the
22 refit began at its own target, and a negative reported from a search that examined
23 zero candidates. This is a reproducibility failure, not evidence that the attributed
24 positions are biologically wrong. We report the conditions under which it is worst,
25 having found none under which the practice is safe, and a reporting checklist.

1 Problem

27 **Biological task.** Given a short RNA sequence and a measured activity, identify *which positions*
28 *determine that activity*. In splice-site and siRNA design this is not a secondary output; it is the
29 object that guides the next round of synthesis. A designer who reads positions 2, 5 and 6 as critical
30 will vary those and hold the rest fixed.

31 **Data modality and setting.** A multiplexed assay of variant effect: each sequence is assayed in
32 a pooled library and read out as a ratio of two sequencing count pools. We use the BRCA2 exon
33 17–19 5′ splice-site assay of Wong et al. [20], which quantifies ~30,483 nine-nucleotide sites with

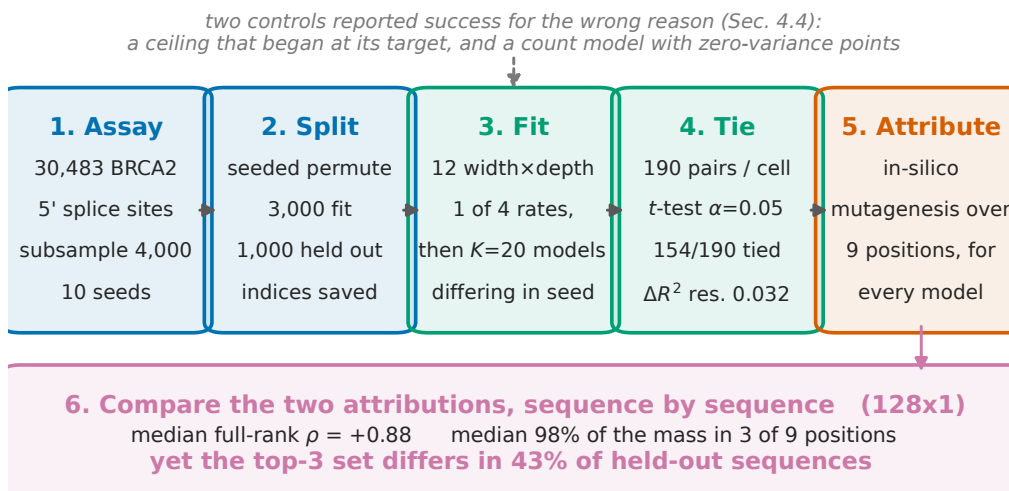


Figure 1: **The pipeline, and where the disagreement appears.** Every count is read from the committed run behind this paper. Models are trained per width $\times$ depth cell differing *only* in initialization seed, then filtered to the pairs a paired t -test on held-out squared errors cannot separate at $\alpha = 0.05$; the surviving pairs are compared by in-silico mutagenesis, sequence by sequence. At the best-fitting cell the survivors agree well on the full-position ranking and concentrate almost all their mass in three positions, yet they disagree about *which* three for nearly half of held-out sequences. The two controls named above the pipeline both reported success without testing what they were meant to test (Section 4, Table 8).

34 $\log_{10}$ percent-spliced-in readouts and retains per-sequence read counts. We subsample $n = 4000$
 35 per seed, split 3000 fit / 1000 held out by a fixed seeded permutation recorded with each run.

36 **Assumption under test.** That the top- k attributed positions recovered from a well-fitting model
 37 are a property of the sequence–activity relationship rather than of the particular model that happened
 38 to be trained.

39 **Target metric.** Agreement of the top- k attributed position *set* between models that *predictive accu-*
 40 *racy* cannot distinguish, measured per-sequence rather than averaged over the library. The qualifier
 41 matters and is the subject of this paper: accuracy-matching is a statement about one aggregate scalar,
 42 not about the functions being close. We use $k = 3$ throughout, justified in Section 4 by the measured
 43 concentration of attribution mass and by Proposition 1 in Appendix A.

44 **Desired outcome.** For the practice to be sound, accuracy-matched models should agree on the top-
 45 k set for nearly all sequences. A designer should not get a different answer because a different seed
 46 was used. If they disagree, either the assay cannot tell the models apart, or it can and the selection
 47 procedure is discarding information the data already holds. We find the second.

48 2 Proposed approach

49 The approach under investigation is standard practice [2, 19], applied in genomics with in-silico
 50 mutagenesis since [21] and still standard [3, 11, 17], not a method we invented: fit a genotype–
 51 phenotype model to the assay, compute per-position attributions by in-silico mutagenesis, and read
 52 the top-ranked positions as the determinants.

53 **Core mechanism.** A model f_θ is fit by minimizing predictive loss on the assay. Attribution scores
 54 each position by the mean absolute change in prediction under substitution, over the real four-letter

55 alphabet:

$$A_j(x) = \frac{1}{3} \sum_{b \neq x_j} |f_\theta(x^{j \rightarrow b}) - f_\theta(x)|. \quad (1)$$

56 Positions are ranked by A_j and the top k reported.

57 **Preconditions.** (i) The model fits the assay well enough that its sensitivities are meaningful. (ii)
 58 The attribution is faithful to the model, in the sense that it reflects the model’s own dependence on
 59 each position [10]. (iii) Implicitly, that the model is *the* model — that the fitting procedure selects a
 60 function determined by the data.

61 **Hypothesis under which it works.** If the data determines f up to predictive equivalence, and
 62 attribution is faithful to f , then the top- k set is determined by the data. Precondition (iii) is the one
 63 we test, and it is the one that is almost never stated.

64 **Models and protocol.** Feedforward genotype–phenotype maps over a grid of widths
 65 $\{16, 32, 64, 128\}$ and depths $\{1, 2, 3\}$, with the per-cell learning rate chosen by pilot on the fit split.
 66 Within a cell we train $K = 20$ models differing *only* in initialization seed: same architecture, same
 67 split, same budget, same learning rate. All reported quantities use at least 5 seeds with bootstrap in-
 68 tervals; enumerations are exact and reported as exact. Claims and thresholds were pre-registered and
 69 committed before any run existed, and their single-commit history predates the earliest run directory
 70 (Appendix C). Total compute is on the order of GPU-days on a single mid-range machine.

71 3 Observed outcome

72 Models that predictive accuracy cannot separate assign different top-3 positions for 43% to 94%
 73 of held-out sequences. They are not, however, hard to tell apart: fewer than three held-out mea-
 74 surements reject that any such pair is the same function. This mirrors the disagreement problem
 75 documented for general-purpose explainers [12], and joins results showing genomic sequence mod-
 76 els can score well while generalizing poorly [16]. **The practice fails by discarding recoverable**
 77 **information.** Selection interrogates one aggregate scalar, and two models can sit inside its resolu-
 78 tion while predicting pointwise differences on the scale of the assay’s noise. Whichever model the
 79 seed produced is then read for attributions as though the data had chosen it.

80 **Accuracy matching.** We filter to pairs whose held-out performance a paired t -test on per-point
 81 squared errors cannot separate at $\alpha = 0.05$. This is a failure to reject, not a demonstration of
 82 equivalence, so we report the filter’s resolution alongside it: the minimum detectable held-out R^2
 83 difference at 80% power given $n_{\text{held-out}} = 1000$. Of 2280 pairs, 1610 are not separable at this
 84 resolution.

85 **Tightening the match does not change the result.** Re-reading the same per-pair records under
 86 stricter caps on the held-out R^2 gap itself, moving from the t -test to $|\Delta R^2| < 0.001$ discards 17
 87 of every 18 surviving pairs and leaves top-3 divergence at 43%, inside the unrestricted interval
 88 (Table 1). The disagreement is not carried by pairs that differ in predictive quality and merely
 89 escaped detection.

90 **It is not an artifact of weak models.** Held-out R^2 does correlate with agreement across cells
 91 (Spearman $+0.650$ against the minimum ρ), so we state the result at the *best*-fitting cell rather than
 92 the worst. At width 128, depth 1 — held-out $R^2 = 0.637$, the highest in the grid — 154 of 190
 93 pairs are accuracy-tied, and among them 37% rank positions at ρ below 0.9, 15% below 0.8, and 6%
 94 below 0.7, reaching $+0.350$ (Figure 2).

95 **The top- k set, which is what gets used.** The top-3 set differs for 73% and 90% of held-out
 96 sequences at width 128, depths 2 and 3; restricted to sequences where the concentration diagnostic
 97 (Section 4) says the comparison is meaningful, 71% and 80%. The conditioned figure carries little
 98 weight at depth 3, where only 11% of sequences at width 128 and 0.3% at width 64 have a well-
 99 defined top-3 under both models.

matching rule	pairs/seed	top-1 differs	top-2 differs	top-3 differs
accuracy-tied (t -test)	154	18%	29%	43% [41, 45]
$ \Delta R^2 < 0.01$	83	18%	29%	42% [40, 45]
$ \Delta R^2 < 0.005$	43	18%	30%	43% [41, 46]
$ \Delta R^2 < 0.001$	9	19%	29%	43% [41, 46]

Table 1: **The disagreement does not come from pairs that merely slipped past a t -test: tightening the accuracy match by a factor of 18 leaves it statistically unchanged.** Best-fitting cell (128×1); each row re-reads the same per-pair records under a stricter cap on the held-out R^2 gap. Entries are the mean over the 10 data-split seeds of that seed’s median over qualifying pairs, bootstrapped over seeds rather than pairs, since pairs within a seed share a split. The top-1 column shows the disagreement is not confined to a negligible third position. All twelve cells: Table 6.

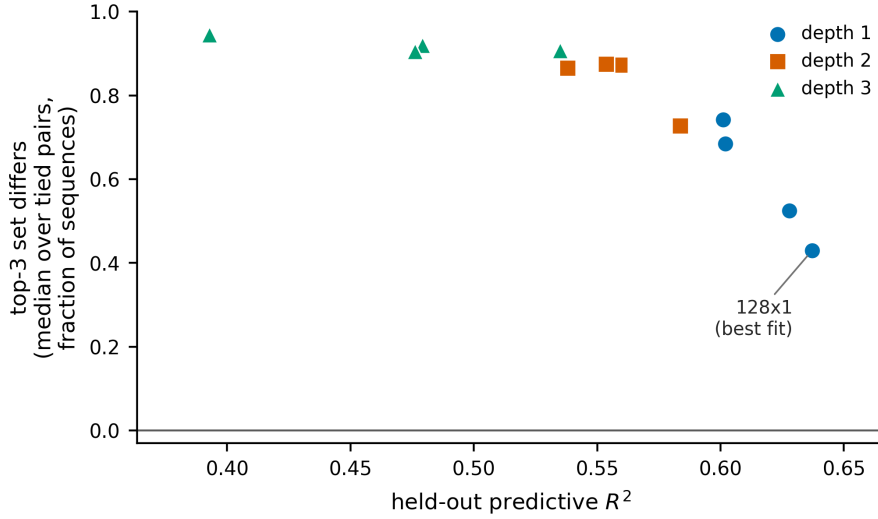


Figure 2: **The failure is not an artifact of poorly-fitting models.** Each point is one architecture cell: held-out predictive R^2 against top-3 attributed-set divergence between accuracy-matched models. **The plotted quantity is the median, over accuracy-tied pairs, of that pair’s fraction of held-out sequences whose top-3 sets differ** — a median over pairs of a per-pair fraction, not a single fraction pooled over sequences; the same definition is used in Tables 3 and 5. The best-fitting cell still diverges on 43% of sequences, so the failure cannot be dismissed as a symptom of underfitting.

cell	held-out R^2	tied pairs	resolution (ΔR^2)	ρ p10 / median	frac $\rho < 0.5$
16×1	0.6011	162/190	0.0699	+0.217 / +0.650	30%
16×2	0.5591	158/190	0.1059	-0.167 / +0.417	58%
16×3	0.5349	134/190	0.1161	-0.300 / +0.283	69%
128×1	0.6372	154/190	0.0318	+0.700 / +0.883	2%
128×2	0.5836	144/190	0.0461	+0.400 / +0.750	16%
128×3	0.4761	95/190	0.0937	-0.267 / +0.317	67%

Table 2: **Models the held-out data cannot tell apart still rank a sequence’s positions differently.** Accuracy-tied pairs and their attribution agreement. A pair is *tied* when a paired two-sided t -test on per-point held-out squared errors does not reject at $\alpha = 0.05$; *resolution* is the smallest held-out R^2 gap that test could have detected at 80% power, so a tie is a failure to reject at a stated resolution rather than a claim of equality. Per-instance ρ is a Spearman correlation between two models’ 9-position attribution vectors *for a single held-out sequence*, pooled over sequences and over tied pairs. **Rows are the two extreme widths at each depth**, which bracket both axes the argument turns on; all twelve cells, with bootstrap intervals, are in Table 5.

cell	top-3 mass	effective positions	top-1 differs	top-3 exact match	mean Jaccard
16×1	0.926	3.03	45%	26%	0.57
16×2	0.819	4.44	60%	13%	0.43
16×3	0.768	5.11	68%	9%	0.37
128×1	0.976	2.20	18%	57%	0.78
128×2	0.918	3.16	36%	27%	0.60
128×3	0.728	5.67	68%	10%	0.38

Table 3: **The disagreement is about which positions carry the mass, not about the ordering of a negligible tail.** Top-3 set agreement and attribution concentration. *top-3 mass* is the median fraction of a sequence’s total $|\Delta|$ in its three strongest positions and *effective positions* is exp of the entropy of the normalized magnitudes, where 9 means all positions contribute equally. Jaccard on 3-element sets takes only $\{0, 0.2, 0.5, 1\}$, so its median is near-useless as a summary and the *mean* is reported, alongside the fraction of sequences whose top-3 sets match exactly. **top-1 differs** is the fraction of sequences on which the two models’ single *strongest* position is a different position — a stronger form of disagreement than a reordering beneath a shared leader, and not implied by the top-3 columns, which are over unordered sets. Rows are the same bracketing subset as Table 2; all twelve cells are in Table 5.

Dominant position, or the order beneath it? A shuffle of ranks 2 and 3 is a far weaker finding than a change of leader, and the top- k Jaccard cannot separate them: it is over unordered sets, so two models can select the same three positions and still disagree about which dominates. We therefore report the strongest position separately (Tables 3 and 1). **The two models pick a different strongest position for 18% of sequences at width 128 depth 1, rising to 68% at depth 3;** the top-2 set differs for 29%, so the disagreement is not confined to a negligible third position. Pooling the 203,200 sequence-pair observations at that cell gives a 44.2% set disagreement, of which 35.0 points agree on the leader and differ only at ranks 2–3. **So at the best-fitting cell most of the top-3 disagreement is about ranks 2 and 3, and we say so rather than claim the stronger result;** at depth 3 the balance inverts, 63.9 of the 91.0 points also involving a different leader. These are accuracy-tied pairs of *independently trained* models, the population described in Section 2. The corresponding figures for a fit against its own reparameterized twin are much smaller — 46%, 47%, 34% and 21% — and are reported separately in Appendix G; the two populations must not be quoted interchangeably.

How cheaply the data adjudicates. Proposition 3 gives the measurements needed to reject that two accuracy-tied models are the same function, under a noise scale derived from the assay’s own read counts (Proposition 2). The requirement is a median of 0.3 to 2.1 in every cell, four orders of magnitude below the 30,483-sequence library (Figure 3); the median pointwise prediction difference runs 0.5 to 2.4 times the observation noise. One held-out measurement would settle which model is right, and selection never takes it. The complementary case — a fit against its own *reparameterized twin*, needing up to 8.8×10^7 measurements yet diverging on top-3 anyway — is a different population, reported in Appendix G.

4 Reason for failure

4.1 Cause 1: the fitting objective does not identify the attribution

Predictive loss constrains f on the observed points. It places no constraint on which of the many equally-accurate functions is selected, and that selection is made by the initialization seed (Remark 1). This is the Rashomon effect [5, 7, 9, 13, 18] instantiated on a real assay, and the same underspecification documented across ML pipelines [6]. What is new here is not that multiplicity exists but its magnitude *at the level of the top- k set*, and — the point this paper turns on — that the multiplicity is *not* forced by the data.

The selection criterion is too coarse, which is a different problem from too little data. The accuracy filter cannot resolve a held-out R^2 gap below 0.032 even at the best-fitting cell, yet two models inside that gap are separable as functions by fewer than three measurements (Section 3). The assay distinguishes them cheaply and the procedure that picked one never asked: the information

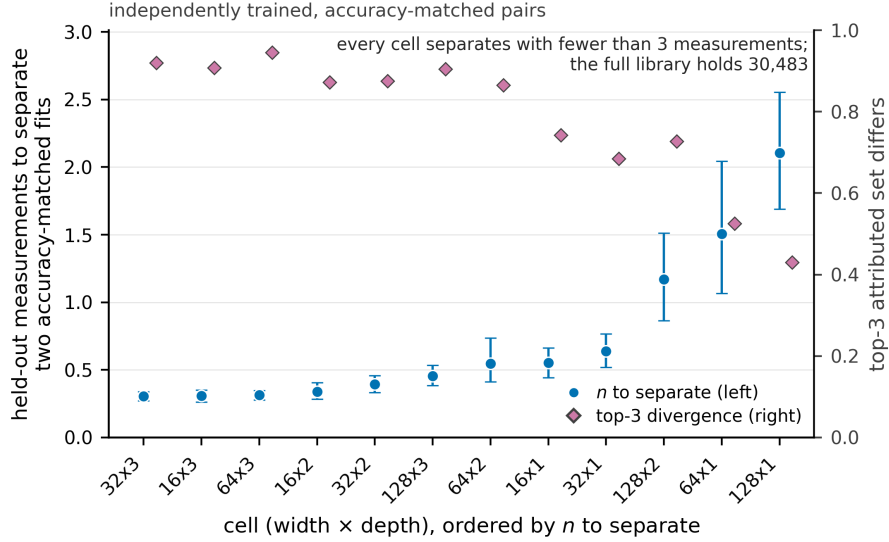


Figure 3: **Accuracy-tied models are cheap to tell apart as functions, yet disagree about which positions matter.** Held-out measurements required to reject that two *independently trained* accuracy-matched models are the same function (Proposition 3), under the assay’s own Poisson count-noise model, with the top-3 set divergence on the right axis. Every cell needs fewer than three measurements, against a library of 30,483, so the reference line is off scale and is stated in the panel rather than drawn. The companion figure for a fit against its own reparameterized twin, where the requirement spans nine orders of magnitude, is Figure 5.

is in the data and the selection step discards it. Had the models been indistinguishable the remedy would be a larger assay; because they are not, it is to compare functions rather than aggregate error (Section 4.6), or to prefer a model interpretable by construction [15].

The consequence is that precondition (iii) of Section 2 is false, and it is the precondition nobody states. Faithfulness of the attribution to the model (precondition ii) does not help: a perfectly faithful attribution of an arbitrarily-selected model is an arbitrary answer.

4.2 Cause 2: attribution is concentrated, so the usual check cannot detect the failure

Attribution is not spread evenly across positions, and how unevenly depends strongly on the cell. The median fraction of a sequence’s $|\Delta|$ in its three strongest positions runs from 54% (width 64 depth 3) to 98% (width 128 depth 1), grid median 86%; effective positions run 7.80 to 2.20 of 9, grid median 3.92 (Table 3, Proposition 4). Figure 4a shows the most concentrated cell, width 128 depth 1: a median 97% in three positions, 0.71 at the single strongest. Quote a per-cell value or a grid median deliberately — they differ by more than ten points here — and match the population, since the twin comparison gives a grid median of 82% and 4.38.

This is not merely a nuisance; it makes the standard check uninformative, and we can say by how much. Attribution is already known to be fragile to small perturbations [8], and there are limits on what any attribution method can certify about a model [4]; the point here is narrower and empirical — the statistic normally used to *check* agreement cannot see the failure. Proposition 1 shows that if two models agree *exactly* on the top 3 of 9 positions and order the remaining 6 at random, the expected Spearman correlation is 0.708. A reported ρ of 0.7 is therefore consistent with perfect agreement about which positions matter, and also with substantial disagreement. The statistic cannot distinguish the two.

Diagnostic. Report the top- k position *set* overlap and the concentration alongside it. Under that diagnostic, ten of twelve cells show genuine disagreement about which positions matter, and it does so in every cell of the grid. Exact top-3 agreement runs 26–57% at depth 1, 12–27% at depth 2 and 6–10% at depth 3. The best cell in the grid, width 128 depth 1, still assigns a different top-3 set

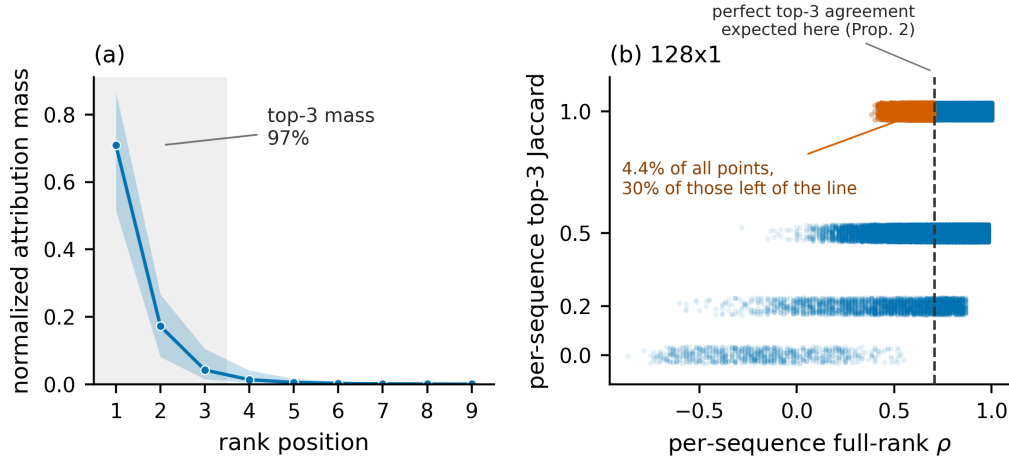


Figure 4: **A low rank correlation has two incompatible readings, and the top- k set separates them.** Accuracy-tied independently trained pairs at width 128, depth 1; three seeds, 203,200 (sequence, pair) points. (a) Normalized attribution mass by rank position, median over 24,000 held-out sequence profiles with interquartile band: a median 97% of the mass sits in the top three positions and the remainder collapses to near zero. (b) Per-sequence full-rank ρ against per-sequence top-3 set Jaccard. The dashed line is $\rho = 0.708$, the value Proposition 1 gives for two models that agree *exactly* on the top 3 and order the remaining 6 at random. Points left of it with Jaccard = 1 (high-lighted) are cases where the two models pick the same three positions while the rank correlation reads as disagreement. **There are 8,861 such (sequence, pair) observations: 4.4% of all 203,200 plotted, and 30% of the 29,310 that fall left of the line.** Each observation is one held-out sequence under one accuracy-tied pair, so the region is a substantial population rather than a handful of outliers. The statistic cannot distinguish the two readings; the top- k set can.

160 for 43% of held-out sequences. We had expected a low-capacity exemption and there is none: the
 161 smallest models are among the *worst*, not the best. See Section 4.5.

162 4.3 Cause 3: the noise model, and a silent failure in ours

163 Asking how much data would separate two models requires a noise scale, and this assay supplies
 164 one rather than needing an assumption. The phenotype is affine in $\log_{10}$ of a count ratio: regressing
 165 y on $\log_{10}(\text{ex}/\text{tot})$ over the library gives slope 0.933 with $r = 0.990$. Those are two independent
 166 count pools, not a proportion — ex exceeds tot in 4.9% of rows, by up to $20.9\times$ — so a binomial
 167 model is wrong here. Proposition 2 gives the Poisson log-ratio scale.

168 **Our first noise model failed silently, in the direction that hides the result.** Under a binomial
 169 form, a near-saturated sequence receives $\sigma \rightarrow 0$; since the separation test of Proposition 3 weights
 170 points by $1/\sigma^2$, a handful of such points dominated and the required sample size collapsed to zero
 171 — that is, the analysis reported that the models were trivially separable. The binomial model is also
 172 inapplicable on its own terms here: $N_{\text{ex}} \geq N_{\text{tot}}$ on 4.9% of rows, where the binomial variance is
 173 zero or negative. Under the Poisson log-ratio the weight distribution is unremarkable — the mean
 174 of $1/\sigma^2$ is 3.0 times its median — and the requirement becomes meaningful. We do not quote a
 175 magnitude for the binomial inflation: that model was replaced before the separation experiment was
 176 first committed, so the figure cannot be regenerated from released code (Appendix F). Counting
 177 noise is a lower bound on total noise, so every reported requirement is a lower bound.

178 4.4 Cause 4: two ways our validation nearly hid the failure

179 Both are errors we committed and caught. We report them because the controls that catch them are
 180 cheap and appear not to be standard, in the spirit of the randomization checks that exposed saliency
 181 methods insensitive to their own model [1]. Table 8, in Appendix D, states each one as what was
 182 reported, what was actually tested, and what the corrected check reports.

183 **A ceiling that was arithmetic, and a negative from an empty search.** To check that a refit
 184 can recover a target, one refits to a target the class provably contains. Ours warm-started from the
 185 reference weights, so for the null target — the reference’s own latent — the initial loss was already
 186 6.7×10^{-19} , below the early-stopping threshold; the loop never ran and the control returned 1.000000
 187 because it *began at the answer*. It verified that the optimizer does not wander away from a perfect
 188 solution, never that the procedure can *find* one it does not start at. Cold-starting the same refit is a
 189 ceiling that requires search, and under it recovery of a target the architecture realizes *exactly* falls to
 190 held-out $R^2 \approx 0.45$ at depth 3 (Appendix D). Separately, an experiment reported “no witness found”
 191 after examining zero candidates, every cell having had an empty candidate set for structural reasons.
 192 Searched-and-found-nothing is a result; never-searched is not, and the second was reported as the
 193 first. We now emit a VOID verdict whenever a negative would be drawn from an empty candidate
 194 set, with a regression test pinning the distinction.

195 4.5 Boundary conditions

196 We found no regime in this grid where the practice is safe. What we can give is the shape of the
 197 degradation, which is measurable and consistent.

- 198 • **Depth is what tracks the failure; width is not, and no cell is safe.** Top-3 divergence
 199 rises monotonically with depth: 43–74% at depth 1, 73–88% at depth 2, 90–94% at depth
 200 3, a rank correlation of +0.92 with depth against −0.28 with width. Within depth 1 it
 201 improves with width, from 74% to 43%, but never falls below 43% anywhere. We looked
 202 for a low-capacity exemption and found the opposite.
- 203 • **Fit quality predicts agreement but does not explain it.** The correlation between held-out
 204 R^2 and agreement is +0.650: real, but far from explanatory, and the best-fitting cell still
 205 shows the failure.
- 206 • **Pairs that are harder to tell apart agree more, not less.** Across cells the median number
 207 of held-out measurements needed to separate a tied pair as functions is rank-correlated
 208 −0.94 with top-3 divergence: where the two fits are closer as functions they also pick more
 209 similar positions, which is the direction one would want and the opposite of what the twin
 210 comparison shows (Appendix G). The failure is therefore not a by-product of the two fits
 211 being nearly identical.
- 212 • **Scope.** One assay, one phenotype, sequence length 9, and each fit draws 4000 of the li-
 213 brary’s 30,483 sequences, so we have not established whether the instability persists at
 214 full-library scale. We tested only in-silico mutagenesis, not gradient or Shapley-based at-
 215 tribution. The result is a reproducibility failure — the same data and procedure return
 216 different top- k sets — not evidence that any particular attributed position is biologically
 217 wrong, which agreement between models cannot settle either way.

218 4.6 Actionable takeaways

- 219 1. **Report per-instance top- k sets, not a library-averaged rank correlation,** which by
 220 Proposition 1 cannot distinguish perfect agreement from substantial disagreement.
- 221 2. **Report the concentration of the attribution.** Without it the reference value for a rank
 222 correlation is unknown, so the correlation cannot be interpreted at all.
- 223 3. **Derive the noise scale from the assay where counts are available,** and check the count
 224 model. Ours was wrong in the direction that hid the problem.
- 225 4. **Train more than one seed and report top- k set agreement across them.** A constant-
 226 factor cost, and the cheapest check that catches this.
- 227 5. **Verify that a passing control required a search,** since one that starts at its own target
 228 passes by arithmetic.
- 229 6. **Refuse to report a negative from an empty candidate set.**

230 References

- 231 [1] Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M. & Kim, B. (2018) Sanity checks for saliency
232 maps. In *Advances in Neural Information Processing Systems 31*. arXiv:1810.03292.
- 233 [2] Alipanahi, B., Delong, A., Weirauch, M.T. & Frey, B.J. (2015) Predicting the sequence specificities of DNA-
234 and RNA-binding proteins by deep learning. *Nature Biotechnology* **33**(8):831–838.
- 235 [3] Avsec, Ž., Weilert, M., Shrikumar, A., Krueger, S., Alexandari, A., Dalal, K., Fropf, R., McAnany, C.,
236 Gagneur, J., Kundaje, A. & Zeitlinger, J. (2021) Base-resolution models of transcription-factor binding reveal
237 soft motif syntax. *Nature Genetics* **53**:354–366.
- 238 [4] Bilodeau, B., Jaques, N., Koh, P.W. & Kim, B. (2024) Impossibility theorems for feature attribution. *Pro-
239 ceedings of the National Academy of Sciences* **121**(2):e2304406120.
- 240 [5] Breiman, L. (2001) Statistical modeling: the two cultures. *Statistical Science* **16**(3):199–231.
- 241 [6] D’Amour, A., Heller, K., Moldovan, D., Adlam, B., Alipanahi, B., Beutel, A., et al. (2020) Underspecifica-
242 tion presents challenges for credibility in modern machine learning. arXiv:2011.03395.
- 243 [7] Fisher, A., Rudin, C. & Dominici, F. (2019) All models are wrong, but many are useful: learning a variable’s
244 importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning
245 Research* **20**(177):1–81.
- 246 [8] Ghorbani, A., Abid, A. & Zou, J. (2019) Interpretation of neural networks is fragile. In *Proceedings of the
247 AAAI Conference on Artificial Intelligence* **33**:3681–3688.
- 248 [9] Hsu, H. & Calmon, F.P. (2022) Rashomon capacity: a metric for predictive multiplicity in classification. In
249 *Advances in Neural Information Processing Systems 35*, pp. 28988–29000.
- 250 [10] Jacovi, A. & Goldberg, Y. (2020) Towards faithfully interpretable NLP systems: how should we define
251 and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational
252 Linguistics*, pp. 4198–4205.
- 253 [11] Koo, P.K. & Eddy, S.R. (2019) Representation learning of genomic sequence motifs with convolutional
254 neural networks. *PLOS Computational Biology* **15**(12):e1007560.
- 255 [12] Krishna, S., Han, T., Gu, A., Wu, S., Jabbari, S. & Lakkaraju, H. (2022) The disagreement problem in
256 explainable machine learning: a practitioner’s perspective. arXiv:2202.01602.
- 257 [13] Marx, C., Calmon, F. & Ustun, B. (2020) Predictive multiplicity in classification. In *Proceedings of the
258 37th International Conference on Machine Learning*, PMLR **119**:6765–6774.
- 259 [14] Novakovsky, G., Dexter, N., Libbrecht, M.W., Wasserman, W.W. & Mostafavi, S. (2023) Obtaining genet-
260 ics insights from deep learning via explainable artificial intelligence. *Nature Reviews Genetics* **24**:125–137.
- 261 [15] Rudin, C. (2019) Stop explaining black box machine learning models for high stakes decisions and use
262 interpretable models instead. *Nature Machine Intelligence* **1**:206–215.
- 263 [16] Sasse, A., Ng, B., Spiro, A.E., Tasaki, S., Bennett, D.A., Gaiteri, C., De Jager, P.L., Chikina, M. &
264 Mostafavi, S. (2023) Benchmarking of deep neural networks for predicting personal gene expression from
265 DNA sequence highlights shortcomings. *Nature Genetics* **55**:2060–2064.
- 266 [17] Seitz, E.E., McCandlish, D.M., Kinney, J.B. & Koo, P.K. (2024) Interpreting cis-regulatory mechanisms
267 from genomic deep neural networks using surrogate models. *Nature Machine Intelligence* **6**:701–713.
- 268 [18] Semenova, L., Rudin, C. & Parr, R. (2022) On the existence of simpler machine learning models. In
269 *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 1827–1858.
- 270 [19] Tareen, A., Kooshkbaghi, M., Posfai, A., Ireland, W.T., McCandlish, D.M. & Kinney, J.B. (2022) MAVE-
271 NN: learning genotype–phenotype maps from multiplex assays of variant effect. *Genome Biology* **23**:98.
- 272 [20] Wong, M.S., Kinney, J.B. & Krainer, A.R. (2018) Quantitative activity profile and context dependence of
273 all human 5’ splice sites. *Molecular Cell* **71**(6):1012–1026.
- 274 [21] Zhou, J. & Troyanskaya, O.G. (2015) Predicting effects of noncoding variants with deep learning-based
275 sequence model. *Nature Methods* **12**:931–934.
- 276 [22] Zhou, S., Huang, M. & Li, K. (2026) Position: genomic model research must move beyond anecdotal eval-
277 uation of interpretability methods. In *Proceedings of the 43rd International Conference on Machine Learning*.
278 arXiv:2606.07607.

279 A Formal statements

280 The statements below are elementary. We give them explicitly because two of them (Propositions 1
281 and 2) determine how the main-text numbers should be read, and because the failure described in
282 Section 4.3 was a violation of Proposition 2 that produced a plausible-looking wrong answer rather
283 than an error.

284 **Remark 1** (Predictive risk does not determine the attribution). *Let $\mathcal{F}$ be a model class, R a popu-*
285 *lation risk, and $\mathcal{F}_\epsilon = \{f \in \mathcal{F} : R(f) \leq \min_{f'} R(f') + \epsilon\}$ the ϵ -Rashomon set. Let $A : \mathcal{F} \rightarrow \mathbb{R}^L$*
286 *be the attribution map of Eq. (1). Nothing in the definition of $\mathcal{F}_\epsilon$ constrains A to be constant on it.*
287 *Consequently, the attribution reported by a fitting procedure is a function of which element of $\mathcal{F}_\epsilon$*
288 *the optimizer returns, and that is determined by the initialization, not by the data. The empirical*
289 *content of this paper is the diameter of $A(\mathcal{F}_\epsilon)$ on a real assay, measured at the level of the top- k set.*

290 **Proposition 1** (Expected rank correlation under concentrated attribution). *Let two models rank L*
291 *positions. Suppose they agree exactly on the ordering of the top k positions, and that the remaining*
292 *$m = L - k$ positions carry negligible attribution mass, so that their relative orderings are inde-*
293 *pendent uniform random permutations. Then the expected Spearman rank correlation over all L*
294 *positions is*

$$\mathbb{E}[\rho_L] = 1 - \frac{m(m^2 - 1)}{L(L^2 - 1)}. \quad (2)$$

295 For $L = 9$ and $k = 3$ this is $1 - 2^{10}/720 = 0.708$.

296 *Proof.* Spearman's ρ over L items is $\rho = 1 - 6 \sum_i d_i^2 / (L(L^2 - 1))$, where d_i is the rank difference
297 at item i . The top k contribute $d_i = 0$ by assumption. For two independent uniform permutations of
298 m items, $\mathbb{E}[\rho_m] = 0$, and since $\rho_m = 1 - 6 \sum d_i^2 / (m(m^2 - 1))$ this gives $\mathbb{E}[\sum d_i^2] = m(m^2 - 1)/6$
299 over the tail block. Substituting yields the result. $\square$

top- k agreed	$k = 2$	$k = 3$	$k = 4$	$k = 5$
$\mathbb{E}[\rho_9]$	0.533	0.708	0.833	0.917

Table 4: Expected Spearman correlation over $L = 9$ positions when two models agree *exactly* on the top k and order the rest at random (Proposition 1). A reported ρ at or below these values carries no evidence of disagreement about which positions matter.

300 **Proposition 2** (Poisson log-ratio noise scale). *Let the phenotype be $y = a \log_{10}(N_{\text{ex}}/N_{\text{tot}}) + b$ with*
301 *N_{ex} and N_{tot} independent Poisson counts with means $\lambda_{\text{ex}}, \lambda_{\text{tot}}$. Then to first order*

$$\text{sd}(y) = \frac{|a|}{\ln 10} \sqrt{\frac{1}{\lambda_{\text{ex}}} + \frac{1}{\lambda_{\text{tot}}}} \approx \frac{|a|}{\ln 10} \sqrt{\frac{1}{N_{\text{ex}}} + \frac{1}{N_{\text{tot}}}}. \quad (3)$$

302 *Proof.* For $N \sim \text{Poisson}(\lambda)$, the delta method with $g(N) = \ln N$ gives $\text{Var}(\ln N) \approx$
303 $\lambda^{-2} \text{Var}(N) = 1/\lambda$. Independence of the two pools gives $\text{Var}(\ln N_{\text{ex}} - \ln N_{\text{tot}}) \approx 1/\lambda_{\text{ex}} + 1/\lambda_{\text{tot}}$.
304 Converting to base 10 and scaling by a yields the result; substituting counts for means is the usual
305 plug-in estimator. $\square$

306 **Remark 2** (Why the binomial form fails here, and fails silently). *Modelling the readout as a propor-*
307 *tion p from N trials gives $\text{sd}(y) \propto \sqrt{(1-p)/(Np)}$, which tends to 0 as $p \rightarrow 1$. Since the statistic*
308 *of Proposition 3 weights points by $1/\sigma_i^2$, a small number of near-saturated sequences then carry*
309 *unbounded weight. Under Proposition 2 the ratio $\text{mean}(1/\sigma^2)/\text{median}(1/\sigma^2)$ is 3.0, a heavy but*
310 *unremarkable tail. We give no corresponding figure for the binomial form: it was replaced before*
311 *the separation experiment was first committed and cannot be regenerated from released code, so*
312 *only the structural defect is quoted. That defect is decisive on its own: $N_{\text{ex}} > N_{\text{tot}}$ in 4.9% of rows,*
313 *by up to $20.9\times$, so the two counts are not numerator and denominator of a proportion.*

314 **Proposition 3** (Separation sample size). *Let f_1, f_2 be two candidate functions with pointwise differ-*
315 *ence $d_i = f_1(x_i) - f_2(x_i)$ at held-out points x_i , and let observations carry independent Gaussian*

noise with known standard deviations σ_i . Testing $H_0 : y \sim f_1$ against $H_1 : y \sim f_2$ at two-sided level α with power $1 - \beta$ requires

$$n \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{(d_i/\sigma_i)^2}, \quad \overline{(d_i/\sigma_i)^2} = \frac{1}{n} \sum_{i=1}^n \frac{d_i^2}{\sigma_i^2}. \quad (4)$$

Proof. The log-likelihood ratio between the two simple hypotheses is Gaussian with mean $\pm \frac{1}{2} \sum_i d_i^2/\sigma_i^2$ under H_0 and H_1 respectively, and variance $\sum_i d_i^2/\sigma_i^2$. The standardized separation is therefore $(\sum_i d_i^2/\sigma_i^2)^{1/2}$, and the standard two-sided power requirement $(\sum_i d_i^2/\sigma_i^2)^{1/2} \geq z_{1-\alpha/2} + z_{1-\beta}$ rearranges to the stated bound. $\square$

Remark 3 (The reported requirement is a lower bound). *Proposition 2 accounts for sequencing count noise only. Library preparation, transfection and biological variation add further variance, which increases σ_i and hence increases the required n . Every separation requirement in the main text is therefore a lower bound on the measurements needed.*

Proposition 4 (Effective number of contributing positions). *For an attribution vector $A \in \mathbb{R}_{\geq 0}^L$ with $\sum_j A_j > 0$, let $q_j = A_j / \sum_{j'} A_{j'}$. The effective number of contributing positions is $N_{\text{eff}}(A) = \exp(-\sum_j q_j \ln q_j)$, satisfying $1 \leq N_{\text{eff}} \leq L$, with $N_{\text{eff}} = L$ iff all positions contribute equally and $N_{\text{eff}} \rightarrow 1$ as one position dominates.*

This is the exponentiated Shannon entropy of the normalized attribution and is reported per-sequence in Table 3; the median over held-out sequences is 3.92 of 9 for accuracy-tied independently trained pairs and 4.38 for the twin comparison, both medians *across cells*; per cell the value ranges from 2.20 to 7.80, so the reference value in Proposition 1 must be read against the cell in question rather than against the grid median.

B Full per-cell results

Table 5 combines Tables 2 and 3 for every cell in the width $\times$ depth grid, with bootstrap intervals on the columns that admit them. It is generated by `experiments/make_tables.py` directly from the committed run directories, as are all tables in this paper.

Table 6 repeats the accuracy-match tightening of Table 1 in every cell. The pair population falls by roughly an order of magnitude and the top- k disagreement does not move anywhere in the grid.

C Protocol and pre-registration

Table 7 gives the per-cell protocol, read from the `tuning_budget.json` written with each run and from the per-seed records. The split rule is the same everywhere: for seed k a generator seeded with $1,000,003 + k$ draws a permutation of the 4000 subsampled sequences, the first 3000 form the fit split and the remaining 1000 are held out, and the indices are recorded in the run directory. All twelve cells use ten seeds.

Architecture and optimizer. The latent map is a fully-connected stack of $\text{depth} \times [\text{Linear}, \tanh]$ followed by a linear projection to a scalar, in `float64`, over a one-hot encoding of the nine-position site. Its output is passed through a MAVE-NN-style monotone global-epistasis nonlinearity: a sum of 5 sigmoids with non-negative weights, hence strictly increasing, fitted jointly with the latent map. All parameters are initialized i.i.d. $\mathcal{N}(0, 0.3^2)$ from the per-model seed, which is the *only* thing that differs between the $K = 20$ models in a cell. Training minimizes mean-squared error on the fit split with Adam, full-batch, for the fixed epoch budget in Table 7, with no weight decay, no dropout and no early stopping; the learning rate is the one chosen by the per-cell pilot. In-silico mutagenesis is evaluated on 400 held-out sequences per pair, so the per-sequence fractions in this paper have that denominator.

Claims, thresholds and go/no-go criteria were written and committed before any result existed: `PREREGISTRATION.md` has a single commit, dated 2026-08-26, and the earliest committed run directory is dated 2026-08-29, so the ordering is checkable from the repository history rather than

(a) accuracy-tied pairs and per-instance agreement					
cell	held-out R^2	tied pairs	resolution	ρ p10 / median	frac $\rho < 0.5$
16×1	0.6011 [0.5725, 0.6274]	162 [156, 169]	0.0699	+0.217 / +0.650	30%
16×2	0.5591 [0.5243, 0.5880]	158 [147, 167]	0.1059	-0.167 / +0.417	58%
16×3	0.5349 [0.4948, 0.5677]	134 [122, 147]	0.1161	-0.300 / +0.283	69%
32×1	0.6019 [0.5754, 0.6263]	167 [160, 172]	0.0651	+0.365 / +0.717	19%
32×2	0.5537 [0.5110, 0.5909]	143 [128, 158]	0.1032	-0.168 / +0.408	58%
32×3	0.4792 [0.4128, 0.5349]	102 [97, 108]	0.1256	-0.317 / +0.258	71%
64×1	0.6278 [0.6003, 0.6544]	160 [151, 169]	0.0402	+0.600 / +0.833	4%
64×2	0.5381 [0.4846, 0.5861]	100 [81, 120]	0.0882	-0.183 / +0.417	57%
64×3	0.3929 [0.3309, 0.4483]	92 [86, 97]	0.1267	-0.383 / +0.175	78%
128×1	0.6372 [0.6122, 0.6585]	154 [138, 166]	0.0318	+0.700 / +0.883	2%
128×2	0.5836 [0.5369, 0.6259]	144 [119, 168]	0.0461	+0.400 / +0.750	16%
128×3	0.4761 [0.3949, 0.5456]	95 [71, 117]	0.0937	-0.267 / +0.317	67%

(b) top- k set agreement and concentration					
cell	top-3 mass	effective positions	top-1 differs	top-3 exact	mean Jaccard
16×1	0.926	3.03	45%	26%	0.57
16×2	0.819	4.44	60%	13%	0.43
16×3	0.768	5.11	68%	9%	0.37
32×1	0.956	2.56	37%	32%	0.62
32×2	0.851	4.05	62%	12%	0.42
32×3	0.759	5.28	69%	8%	0.36
64×1	0.969	2.36	24%	48%	0.73
64×2	0.873	3.79	61%	14%	0.43
64×3	0.540	7.80	74%	6%	0.31
128×1	0.976	2.20	18%	57%	0.78
128×2	0.918	3.16	36%	27%	0.60
128×3	0.728	5.67	68%	10%	0.38

Table 5: **No cell is withheld: the pattern in Tables 2 and 3 holds across the whole grid.** All twelve cells. The upper panel carries the columns of Table 2, the lower those of Table 3. **Which columns carry intervals, and why the rest do not:** held-out R^2 and the tied-pair count are per-seed quantities and are given as mean with a 95% bootstrap interval over the ten seeds. The remaining columns are medians pooled over every tied pair across all seeds rather than per-seed values, so a seed-level bootstrap interval is not defined for them and none is invented here.

width	depth	accuracy-tied (t -test)				$ \Delta R^2 < 0.001$			
		pairs	top-1	top-2	top-3	pairs	top-1	top-2	top-3
16	1	162	45%	62%	74% [73, 76]	6	43%	60%	72% [68, 76]
16	2	158	60%	79%	87% [86, 89]	3	58%	77%	86% [83, 88]
16	3	134	67%	84%	90% [88, 92]	2	65%	82%	88% [84, 91]
32	1	167	36%	55%	68% [67, 70]	5	35%	52%	66% [63, 69]
32	2	143	59%	80%	87% [86, 89]	3	62%	81%	88% [86, 90]
32	3	102	68%	86%	92% [91, 93]	1	69%	86%	92% [91, 94]
64	1	160	24%	38%	52% [47, 57]	6	24%	37%	49% [44, 54]
64	2	100	57%	79%	86% [82, 89]	2	53%	76%	85% [80, 90]
64	3	92	73%	89%	94% [94, 95]	1	67%	84%	90% [82, 94]
128	1	154	18%	29%	43% [41, 45]	9	19%	29%	43% [41, 46]
128	2	144	38%	61%	74% [71, 77]	6	29%	55%	72% [68, 76]
128	3	95	68%	86%	91% [89, 92]	2	63%	85%	92% [89, 94]

Table 6: Tightening the accuracy match to $|\Delta R^2| < 0.001$ in every cell. The pair population falls by roughly an order of magnitude and the top- k disagreement does not move, so the failure is not an artifact of the t -test’s resolution. Columns as in Table 1; intervals bootstrap over data-split seeds.

360 asserted. Amendment is additionally blocked by a pre-write hook in the authoring environment
361 that matches the target path of a write; we name the mechanism precisely because it constrains our

cell	learning rate chosen (seeds)	configs	epochs	grad. steps
16×1	0.01	4	1,200	1,200
16×2	0.01	4	1,200	1,200
16×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
32×1	0.01	4	1,200	1,200
32×2	0.01	4	1,200	1,200
32×3	0.01	4	1,200	1,200
64×1	0.01 (×4), 0.003 (×6)	4	1,200	1,200
64×2	0.01 (×9), 0.003 (×1)	4	1,200	1,200
64×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
128×1	0.003 (×9), 0.001 (×1)	4	1,200	1,200
128×2	0.003 (×4), 0.001 (×6)	4	1,200	1,200
128×3	0.003 (×4), 0.001 (×6)	4	1,200	1,200

Table 7: **The tuning budget is four learning rates per cell and nothing else, so the disagreement cannot be an artifact of unequal search.** Per-cell protocol, read from `tuning_budget.json` and the per-seed records. The search space is “learning rate over [0.01, 0.003, 0.001, 0.0003], selected once per cell and shared by all models in it”. The selection rule is identical in every cell — “no model selection: every trained model is kept and enters the pairwise analysis” — so it is stated here rather than repeated as a column. **The chosen rate is not constant across seeds:** it varies in 7 of 12 cells, and the multiplicities are given. Within any one seed a single rate is shared by all 20 models in the cell, so the models still differ only in initialization; across seeds the rate is re-selected.

check	what it reported	what it actually tested	corrected control
Ceiling control	Held-out $R^2 = 1.000000$ recovering a target the architecture realizes exactly, in every cell and every seed.	That Adam does not leave a perfect solution. Warm-started <i>at</i> the target, its initial loss was already below the stopping threshold and best-iterate selection kept the initial state, so the loop never ran.	Cold-started on the same target, the worst cell reaches only 0.4501 and 11 of 12 cells fall below the 0.999 threshold.
Noise model	A required sample size of zero: any two fits separable with no data at all.	Inverse-variance weighting under a count model undefined on this library: the pools are not a proportion, $\text{ex} \geq \text{tot}$ on 4.9% of rows, so binomial variance is zero or negative there and a few near-saturated sequences carried the statistic.	Under the Poisson log-ratio the weights are ordinary (mean $1/\sigma^2$ is $3.0\times$ its median) and the requirement is finite, 0.5 to 87,751,464 measurements.

Table 8: **Both controls we relied on were passing for reasons unrelated to what they were meant to test.** Neither failure was visible from the reported number alone: a control that cannot fail and a weighting scheme that silently concentrates on a few points both look like success.

362 workflow rather than the repository, and the durable evidence is the single-commit history. Every
363 reported number carries the git SHA and configuration hash of the run that produced it.

364 D The vacuous ceiling, in full

365 The refit loads the reference weights, then evaluates the loss and snapshots the best iterate *before*
366 entering the optimization loop. For the null target — the reference’s own latent — that initial state
367 is already the best iterate that will ever be seen, so no later step can displace it. The measured
368 initial loss is 6.7×10^{-19} , below the 10^{-10} early-stopping threshold, and the loop breaks on its first
369 iteration.

370 The consequence is visible in the run records without reference to any loss value: the warm-started
371 control returns held-out R^2 of exactly 1.0 in all 120 cell–seed combinations, bit-exact rather than
372 approximate. A control that cannot return anything else has not measured anything. The cold-started
373 replacement, which must actually travel to the same in-class target, recovers it at only 0.4501 in the

374 worst cell and falls below the 0.999 threshold in 11 of 12 cells, with per-seed spread of 0.21 to 0.87
 375 at width 128 depth 3. Both arms are plotted in Figure 6.

376 The general form of the error is worth stating separately from its instance. A control that verifies a
 377 procedure can recover a target must ensure the procedure does not *begin* at that target. Warm-starting
 378 a refit from the reference weights and then asking it to recover the reference’s own output satisfies
 379 the letter of the control and none of its purpose.

380 E Reproducibility

381 Code, pre-registered analysis plans, per-seed raw values and run provenance will be released on
 382 acceptance; the repository is withheld here for anonymity. Each result directory records the git
 383 SHA, the configuration hash, the seed list and the tuning budget consumed, and every table and
 384 figure in this paper is generated from those directories by scripts included in the release rather than
 385 transcribed by hand.

386 F Noise model derivation

387 Z requires an observation-noise scale. This library supplies one rather than needing an assump-
 388 tion, and the derivation is given here because the choice of count model changes Z by orders of
 389 magnitude.

390 **The phenotype is affine in a log count ratio.** Each sequence carries two count pools, *ex*
 391 and *tot*. Regressing the phenotype y on $\log_{10}(\text{ex}/\text{tot})$ over the full library gives slope 0.9331
 392 [0.9314, 0.9345] across seeds, and $r = 0.990$ over the full 30,483-sequence library. The relation-
 393 ship is tight enough that the count ratio, not the phenotype, is the quantity whose noise must be
 394 modelled.

395 **Why Poisson and not binomial.** The two pools are *not* numerator and denominator of a propor-
 396 tion. Over the committed library, *ex* exceeds *tot* in 4.9% of rows, with ratios as large as 20.9, and
 397 $\text{ex} = 0$ in 11.4% of rows. A binomial model is therefore not merely inefficient but undefined on
 398 roughly one row in twenty: $p = \text{ex}/\text{tot} \geq 1$ there, so the binomial variance $p(1 - p)/\text{tot}$ is zero or
 399 negative.

400 **Delta method.** Treating both pools as independent Poisson variates and applying the delta method
 401 to the log of their ratio,

$$\text{var}(\log_{10} R) = \frac{1}{\ln(10)^2} \left(\frac{1}{\text{ex}} + \frac{1}{\text{tot}} \right), \quad \text{sd}(y_i) = |a| \frac{\sqrt{1/\text{ex}_i + 1/\text{tot}_i}}{\ln 10}, \quad (5)$$

402 with a the fitted slope and a half-count continuity correction applied to both pools, which also covers
 403 the rows with $\text{ex} = 0$. In the standardized units the models see, the median of this quantity is 0.3114
 404 [0.3051, 0.3194].

405 **The diagnostic that surfaced the binomial failure.** The separation test weights points by $1/\sigma_i^2$,
 406 so a point assigned near-zero noise dominates the statistic. Under a binomial model a near-saturated
 407 sequence receives exactly that. The symptom was a weight distribution with a mean far above its
 408 median and a required sample size that collapsed to zero for every cell — an implausible answer
 409 that is what prompted the check. Under the committed Poisson log-ratio model the mean of $1/\sigma^2$ is
 410 3.0 times its median, a heavy but unremarkable tail, and Z becomes finite and cell-dependent. An
 411 unexamined count model would have reported that no data at all is needed to separate the fits.

412 We deliberately do not quote a magnitude for the binomial inflation. That model was replaced before
 413 the first commit of the separation experiment, so its exact variance form is not in the repository and
 414 the figure cannot be regenerated from committed code. What *is* reproducible from the committed
 415 library is the structural defect above — $p \geq 1$ on 4.9% of rows, hence zero or negative binomial
 416 variance — together with the Poisson ratio of 3.0. Both were re-derived from the committed 30,483-
 417 row library while preparing this appendix and agree with the recorded values.

418 *A figure comparing the two weight distributions was considered and cut.* It would need the bino-
419 mial variance form, which is not in the repository, and per-observation $1/\sigma^2$ values, which no run
420 directory stores — only the median noise scale survives into `results.json`. Drawing it would
421 have meant inventing the comparator and recomputing the rest outside any recorded run, so the
422 comparison stays in prose where its provenance can be stated.

423 **Z is a lower bound.** This models sequencing noise only. Library preparation and biological vari-
424 ation add more, so the true noise is larger and the required sample size larger still. Every Z in this
425 paper should be read as “at least this many”.

426 G Separability against a reparameterized twin

427 The main text measures *independently trained* models that predictive accuracy cannot separate, and
428 shows that the assay itself separates them cheaply — fewer than three held-out measurements. This
429 section reports the complementary case: a single fit against a *reparameterized twin* of itself, where
430 the data genuinely cannot adjudicate.

431 **The two cases bracket the problem.** In the main-text population the divergence is avoidable: one
432 measurement would settle which of the two models is right, and the selection procedure simply does
433 not ask. Here it is not avoidable — at width 128 depth 3 the twin needs 8.8×10^7 measurements,
434 more than three orders of magnitude beyond the full library — and the top-3 sets diverge regard-
435 less. Attribution therefore fails in both regimes, for different reasons, and only the first has a cheap
436 remedy.

437 The two sets of numbers are not interchangeable. The twin comparison finds substantially more
438 top-3 agreement than independently trained pairs do, so quoting a twin figure for the main-text
439 population would understate the failure.

440 Run 20260830T030657Z_972496a_5da8cb38, git 972496a, config 5da8cb38, 10 seeds, se-
441 quence length 9.

width	depth	held-out R^2	per-instance median ρ	Z to separate
16	1	0.984087 [0.979785, 0.987518]	0.892 [0.877, 0.903]	1 [0, 1]
16	2	0.999241 [0.998708, 0.999619]	0.978 [0.973, 0.983]	42 [25, 60]
16	3	0.999583 [0.999138, 0.999916]	0.984 [0.978, 0.990]	313 [87, 571]
32	1	0.992155 [0.990567, 0.993674]	0.872 [0.847, 0.893]	0 [0, 1]
32	2	0.997328 [0.995451, 0.999059]	0.915 [0.892, 0.937]	87 [14, 193]
32	3	0.999166 [0.998775, 0.999529]	0.923 [0.898, 0.945]	833 [18, 2,136]
64	1	0.998324 [0.997555, 0.998958]	0.899 [0.886, 0.913]	2 [1, 3]
64	2	0.999997 [0.999995, 0.999998]	0.855 [0.832, 0.880]	4,176 [1,320, 7,333]
64	3	0.999996 [0.999993, 0.999998]	0.847 [0.835, 0.860]	7,300 [2,156, 14,436]
128	1	0.999876 [0.999862, 0.999889]	0.923 [0.913, 0.933]	9 [7, 11]
128	2	1.000000 [1.000000, 1.000000]	0.905 [0.888, 0.920]	3,158,529 [1,402,633, 5,004,083]
128	3	1.000000 [1.000000, 1.000000]	0.890 [0.885, 0.895]	87,751,464 [39,140,215, 138,392,840]

Table 9: **A fit and its own twin are far closer than two independently trained fits, which is why the two populations must not be quoted interchangeably.** Z is the number of held-out measurements needed to reject that the two are the same function at $\alpha = 0.05$ with power 0.8, under the noise model of Appendix F. It is a lower bound: it counts sequencing noise only.

442 H Cold-start recovery, per cell

443 **The defect.** The held-out closure experiment (run 20260829T181937Z_b2951ae_9fdd4067, git
444 b2951ae) reported that no cell contains the reparameterized twin, on the strength of a ceiling control
445 that does not do the work it appears to do. Its refit path, `closure_heldout.refit_on_split`,
446 loads the reference weights, then evaluates the loss and snapshots the best iterate *before* entering the
447 optimisation loop. For the null target — which is the reference’s own latent — the refit therefore
448 begins at the answer. The measured initial loss is 6.7×10^{-19} , already below the 10^{-10} early-stop
449 tolerance, so the loop breaks on its first iteration and returns the reference unchanged.

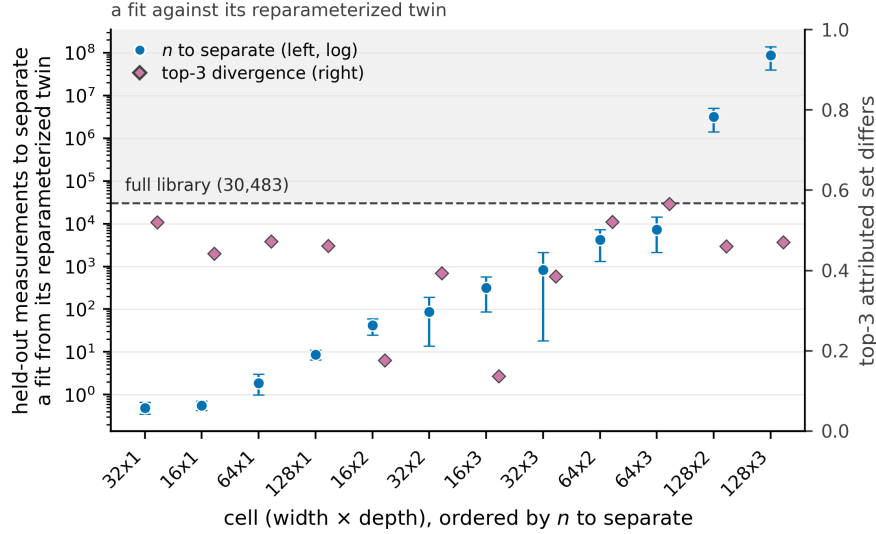


Figure 5: **Separability and top-3 divergence for a fit against its own reparameterized twin, not for independently trained pairs.** The requirement spans nine orders of magnitude here and two cells exceed the full library, where the main-text population separates with fewer than three measurements in every cell (Figure 3). Left axis logarithmic; the dashed line is the full library.

450 A held-out R^2 of 1.000000 on that target is arithmetic, not evidence. It establishes that Adam does
 451 not wander away from a perfect solution. It does not establish that the refit can *find* a solution it does
 452 not start at, which is the capability the containment conclusion depends on. Best-iterate selection
 453 is what makes the failure silent: because the initial state is already the best iterate ever seen, no
 454 subsequent step can displace it, and the procedure reports a perfect score without having searched.

455 **The control that replaced it.** Run 20260829T191427Z_94dd5a2_2267b74e, git 94dd5a2, con-
 456 fig 2267b74e, 10 seeds. The null target is refit *without* warm-starting from the reference, from a
 457 cold random initialization. The target is known to be in the class, since the reference realises it, so
 458 this asks only whether the optimiser can travel to it. Mean initial loss across cells confirms the refit
 459 no longer begins at the answer.

width	depth	cold-start null, held-out R^2	warm warp refit, held-out R^2
16	1	0.9988 [0.9982, 0.9993]	0.9748 [0.9669, 0.9809]
16	2	0.9476 [0.9244, 0.9703]	0.9935 [0.9895, 0.9962]
16	3	0.7763 [0.6820, 0.8589]	0.9959 [0.9921, 0.9986]
32	1	0.9994 [0.9990, 0.9997]	0.9749 [0.9708, 0.9789]
32	2	0.8258 [0.7303, 0.9103]	0.9463 [0.9185, 0.9724]
32	3	0.5626 [0.4404, 0.6891]	0.9404 [0.9151, 0.9643]
64	1	0.9971 [0.9955, 0.9984]	0.9804 [0.9696, 0.9893]
64	2	0.7122 [0.5620, 0.8561]	0.9236 [0.8929, 0.9532]
64	3	0.4501 [0.3529, 0.5575]	0.9034 [0.8879, 0.9186]
128	1	0.9969 [0.9962, 0.9975]	0.9885 [0.9868, 0.9900]
128	2	0.8997 [0.8316, 0.9529]	0.9575 [0.9381, 0.9734]
128	3	0.5328 [0.4000, 0.6714]	0.9247 [0.9108, 0.9394]

Table 10: The ceiling that requires search. The cold-start column is an upper bound on what this procedure can say about any target, because its target is in the class by construction. Mean cold-start initial loss ranges from 1.826 to 2.093, against 6.7×10^{-19} for the warm-started control it replaces.

460 **What the control shows.** Read by depth: at depth 1 the worst-cell cold-start null reaches 0.9969
 461 against 0.9748 for the warm warp refit, so the ceiling sits above the measurement; at depth 2 the
 462 worst-cell cold-start null reaches 0.7122 against 0.9236 for the warm warp refit, so the ceiling sits

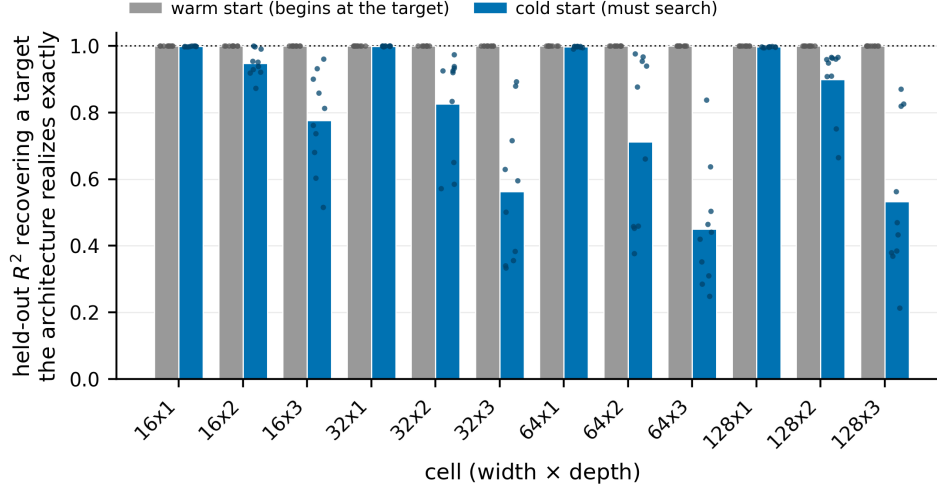


Figure 6: **The warm-start control cannot fail, so it measures nothing.** Held-out R^2 recovering the null target, a function the architecture realizes exactly, with the ten per-seed values overlaid. The warm-started control returns 1.000000 in every cell and every seed — all 120 values are exactly 1.0, not approximately — because it is initialized at the reference and its initial loss of 6.7×10^{-19} is already below the 10^{-10} stopping threshold, so the optimization loop never runs and best-iterate selection keeps the initial state. The cold-started arm must actually search for the same target and its per-seed spread shows how much the basin varies: at 128×3 restarts land between 0.21 and 0.87. Warm bars are flat by construction; that flatness is the finding, not a plotting artefact.

below the measurement; at depth 3 the worst-cell cold-start null reaches 0.4501 against 0.9034 for the warm warp refit, so the ceiling sits below the measurement.

At depth 1 the ceiling sits above the measurement, so the comparison is valid there and the warp target is genuinely not reached as well as an in-class one. At depths 2 and 3 the ranking inverts: the warm-started warp refit scores *higher* than a cold-started refit to a target guaranteed to be in the class. That is initialization dominating the result, and no conclusion about what the class contains can be drawn at those depths. The configured pass threshold is 0.999, and 11 of 12 cells fall below it.

The honest reading is therefore narrower than the original claim: the refit procedure did not find a member of the class equal to the twin. That is consistent with no such member existing, and equally consistent with one existing that the optimiser failed to reach.